Assignment 3- MERN Lab (Program 7)

7. Write a react program to display Hello World using automate both JSX compilation and server restarting

Steps:

Create a Project: Automate

cd automate

npm init –y

npm install express //Install Express

npm install --save-dev @babel/core @babel/cli @babel/preset-react //Install Babel

npm install --save-dev nodemon

package.json

Add 3 lines inside the scripts

"scripts": {

"start": "nodemon -w server server/server.js",

"compile": "babel src --presets @babel/preset-react --out-dir public",

"watch": "babel src --presets @babel/preset-react --out-dir public --watch --verbose"

}

Server.js

```
const express = require('express');
const app = express();
app.use(express.static('public'));
app.listen(3000, function () {
   console.log('Server started at http://localhost:3000');
});
```

First compile the JSX

- npm run compile

Start automatic JSX compilation

- npm run watch

Open a **second terminal**.

- Make sure you are in the project directory:

- npm start
- You should get: Server started at http://localhost:3000
- Open: http://localhost:3000

```
program7
│
├── src
│   └── App.jsx
│
├── public
│   ├── index.html
│   └── App.js          ← generated automatically
│
├── server
│   └── server.js
│
└── package.json
```

Answer:

public/App.js

```
const element = /*#__PURE__*/React.createElement("h1", null, "Hello World");

ReactDOM.render(element, document.getElementById("root"));
```

public/index.js

```
<!DOCTYPE html>

<html>

<head>

  <title>Hello World</title>

</head>

<body>

  <div id="root"></div>


  <script src="https://unpkg.com/react@17/umd/react.development.js"></script>
```

```
  <script
src="https://unpkg.com/react-dom@17/umd/react-dom.development.js"></script>
  <script src="App.js"></script>
</body>
</html>
```

server/server.js

```
const express = require('express');
const app = express();
app.use(express.static('public'));
app.listen(3000, function () {
  console.log('Server started at http://localhost:3000');
});
```

src/App.jsx

```
const element = <h1>Hello World</h1>;
ReactDOM.render(
  element,
  document.getElementById("root")
);
```

Output:



